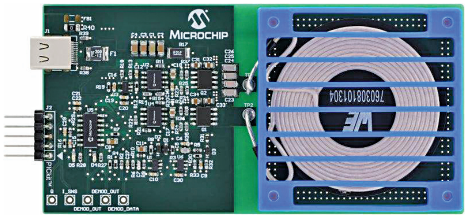
Interesting Reference Designs Of

Title of Reference Design	5-Watt Wireless Charger	15-Watt Wireless Charging Receiver
Image		
Key Attractions	This reference design for a Wireless 5W Charging Transmitter offers a 70% efficiency rating. The design runs on 5V and is compatible with 2A phone chargers. The design includes foreign object detection (FOD) mechanisms for safe operation. The operating frequency ranges from 110kHz to 205kHz.	This reference design for a 15-Watt Buck Wireless Charging Receiver complies with the Wireless Power Consortium (WPC) and Medium Power Working Group (MPWG) specifications. The design converts 3.5-20V AC to 15W, giving 5V at 3A.
Highlights	The AN4880 from Microchip is a reference design for a 5W wireless charging transmitter. The system boasts an overall efficiency exceeding 70%, supports Li-ion charging algorithm, customisable for different chemistries. The design is powered by a 5V USB-C connector, compatible with 2A phone chargers. The transmitter circuit is made up of a microcontroller, two MOSFET drivers, and three operational amplifiers. The wireless charging transmitter operates within a 5W power limit. The design uses efficiency-based FOD to prevent unintended charging, ensuring safety. The operating frequency ranges from 110kHz to 205kHz. The design of the transmitter includes protection against voltage and current fluctuations during operation. The reference design features a user interface connected via a universal asynchronous receiver transmitter (UART). The design empowers users to customise the settings according to their specific needs by allowing configurable receiver detection and protection trigger thresholds. Adjustable FOD trigger thresholds offer flexibility in managing foreign object detection.	WPR1500-BUCK is a reference design for a 15-watt buck wireless charging receiver from NXP. The design complies with the Wireless Power Consortium (WPC) and Medium Power Working Group (MPWG) specifications and can support forthcoming standards. The design's receiver coil accepts input power from 3.5V to 20V AC, converting it efficiently into 15W output power. It supports frequency-shift keying (FSK) communication from medium power transmitters. Hardware protection ensures system safety, covering rectifier voltage, output voltage, and output current. The design achieves space-efficient integration with a compact 40mm×40mm printed circuit board (PCB) size. The reference design is based on the ARM Cortex M0+ processor with ultra-high voltage (UHV) technology. The rectifier in this design utilises a self-driven sync type and has an input voltage range from 3.5V to 20V AC at its peak, while the output voltage ranges from 3.5V to 20V DC. The design includes a DC-DC converter that accepts input voltage from 5V to 21V DC and provides a 5V DC output with a 3A current capacity.
Applications	The reference design is suitable for phone chargers, Qi enabled devices, etc.	The reference design is suitable for various applications such as smartphone chargers, wearable devices, IoT devices, medical devices, etc.
OEM Brand	Microchip	NXP
URL	https://www.electronicsforu.com/electronics-projects/reference-design-for-5-watt-wireless-charger	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-15-watt-wireless-charging-receiver